JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 7 – Biodiversity & Wildlife Conservation

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Biodiversity & Wildlife Conservation** — what biodiversity means, why India's wildlife matters globally, and the practical ways every citizen can support conservation. Whether you are a student, a nature enthusiast, a farmer, or a civic volunteer, this handbook is written so you can read it once and walk away confident and informed.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of biodiversity and wildlife conservation in India.

Learning Objectives

After completing this module, you will be able to:

- Define biodiversity and explain why it matters at multiple levels
- Identify India's biodiversity hotspots and protected area categories
- Explain the difference between national parks, wildlife sanctuaries, and biosphere reserves
- Understand the causes and consequences of habitat loss and human-wildlife conflict
- Recognise India's key conservation success stories, such as Project Tiger
- Understand the government bodies and laws protecting India's wildlife
- Apply practical, citizen-level conservation habits
- Know exactly how and where to report wildlife crime

Why This Topic Matters

India is one of the world's 17 "megadiverse" countries, home to nearly 7% of the world's plant species and a similarly outsized share of global wildlife, despite covering just over 2% of the world's land area. This natural wealth supports agriculture, medicine, water security, and climate resilience for over a billion people.

> ***Reality Check:*** *India's protected area network has grown to 1,014 protected areas — 106 national parks, 573 wildlife sanctuaries, 115 conservation reserves, and 220 community reserves — covering about 5.32% of the country's geographical area (MoEFCC, 2023). India's tiger population has also recovered from a low of 1,411 in 2006 to 3,682 in 2023, and the country is now home to roughly 70–75% of the world's remaining wild tigers.*

Yet this progress remains fragile. The Great Indian Bustard, for instance, has been reduced to fewer than 150 individuals in the wild, threatened largely by collisions with power lines — a reminder that conservation success in one species does not guarantee safety for another.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential module.

Introduction

Biodiversity refers to the variety of life on Earth at every level — genetic, species, and ecosystem — encompassing all plants, animals, fungi, and microorganisms and the ecological complexes they form. **Wildlife conservation** is the practice of protecting wild species and their habitats from decline, extinction, and human-caused harm.

In simple words: biodiversity is the full richness and variety of life around us, and conservation is the deliberate effort to keep that richness intact for the ecosystems and people who depend on it.

India's biodiversity is exceptional by global standards — the country is one of only 17 “megadiverse” nations and hosts four of the world's recognised biodiversity hotspots, making its conservation choices globally significant, not just locally important.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of biodiversity is built upon.

Levels of Biodiversity

Biodiversity operates at three interconnected levels, each important for ecological resilience.

Level	What It Means
Genetic Diversity	Variation within a single species, which builds resilience to disease and environmental change
Species Diversity	The variety of different species within a habitat or region
Ecosystem Diversity	The variety of habitats and ecological communities across a landscape

Key Definitions

- **Megadiverse Country** — One of a small group of nations, including India, home to the majority of the world's species.
- **Biodiversity Hotspot** — A region with exceptional species richness that is also significantly threatened by habitat loss.
- **Endangered Species** — A species facing a very high risk of extinction in the wild, as classified on the IUCN Red List.
- **Habitat Fragmentation** — The breaking up of large, continuous habitats into smaller, isolated patches.
- **Human-Wildlife Conflict** — Negative interactions between people and wild animals, usually arising from competition over shared space and resources.

Complete Detailed Explanation

Part A — India's Protected Area System, In Depth

India protects its biodiversity through a tiered system of designated areas, each with a different purpose and level of restriction on human activity.

Category	Purpose
National Park	Highest protection level; no human settlement or resource extraction allowed
Wildlife Sanctuary	Protects specific species/habitats; limited regulated human activity permitted
Conservation Reserve	Buffer zones adjoining protected areas, often on government land
Community Reserve	Areas managed with community participation, often on private or community land
Biosphere Reserve	Balances conservation with sustainable use by local communities, recognised internationally under UNESCO's MAB programme

Part B — The Ten Building Blocks of Module 7

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian citizens.

1. What is Biodiversity?

Biodiversity refers to the variety of life on Earth at every level — genetic, species, and ecosystem — encompassing all plants, animals, fungi, and microorganisms and the ecological complexes they form.

2. Importance of Biodiversity

Biodiversity underpins essential ecosystem services such as pollination, water purification, soil fertility, and climate regulation. It also provides food, medicine, and livelihoods, and strengthens ecosystem resilience to disturbances and disease.

3. Wildlife Protection

Wildlife protection involves safeguarding animal species and their habitats from poaching, illegal trade, habitat destruction, and human-caused decline, through legal protection, patrolling, and habitat restoration.

4. Forest Conservation

Forests regulate climate, prevent soil erosion, support water cycles, and house most of India's terrestrial biodiversity. Conservation involves controlling deforestation, promoting sustainable practices, and protecting forest ecosystems from encroachment.

5. National Parks & Wildlife Sanctuaries

India's protected area network spans 1,014 sites covering about 5.32% of the country's geographical area, ranging from strictly protected national parks to community-managed reserves, conserving critical habitats across diverse ecosystems.

6. Endangered Species

Endangered species are those at high risk of extinction due to habitat loss, poaching, climate change, or pollution. Conservation status is tracked globally through frameworks like the IUCN Red List to prioritise protection efforts.

7. Afforestation & Reforestation

Afforestation involves planting trees in areas without prior forest cover, while reforestation restores previously forested land. Both combat habitat loss, sequester carbon, and rebuild wildlife corridors.

8. Human-Wildlife Conflict

As human settlements expand into natural habitats, conflicts arise over resources and safety — crop raiding, livestock predation, and accidents. Mitigation includes buffer zones, early warning systems, and community-based coexistence strategies.

9. Conservation Laws

India's legal framework, anchored by the Wildlife Protection Act, 1972 and the Forest (Conservation) Act, 1980, establishes the rules and penalties that protect wildlife and natural habitats nationwide.

10. Citizen Participation

Citizens can support biodiversity conservation by reporting illegal wildlife trade, supporting local afforestation drives, avoiding products from endangered species, participating in citizen-science wildlife surveys, and creating pollinator-friendly spaces.

Important Terminologies

Term	Simple Meaning
Biodiversity Hotspot	A region with exceptional species richness that is also significantly threatened
Core Zone	The strictly protected inner area of a tiger reserve, off-limits to human activity
Buffer Zone	The surrounding area of a protected reserve, allowing regulated sustainable use
Endemic Species	A species found naturally in only one specific geographic region
Habitat Fragmentation	The breaking up of large, continuous habitats into smaller, isolated patches
IUCN Red List	The global inventory of conservation status for species, from Least Concern to Extinct
Keystone Species	A species that has a disproportionately large effect on its ecosystem relative to its abundance
Megadiverse Country	One of a small group of nations home to the majority of the world's species
Poaching	The illegal hunting or capturing of wild animals
Wildlife Corridor	A protected passage linking separated habitats, allowing animals to move safely between them

Step-by-Step Understanding

How Project Tiger Rebuilt India's Tiger Population

1. Crisis Recognition — The 1972 tiger census revealed a crisis: only 1,800 tigers remained in India

2. Legal Protection — The Wildlife Protection Act, 1972 banned tiger hunting nationwide

3. Reserve Creation — Project Tiger launched in 1973 with 9 dedicated reserves, since expanded to 58

4. Habitat Protection — Core zones were kept free of human activity, with voluntary, compensated village relocation

5. Scientific Monitoring — Camera traps, pug-mark surveys, and DNA analysis enabled accurate population tracking every four years

6. Recovery — The population grew from a low of 1,411 in 2006 to 3,682 in 2023

How to Support Wildlife Conservation as a Citizen

1. Report suspected poaching or illegal wildlife trade to the forest department or wildlife crime helpline

2. Avoid purchasing products made from wildlife parts, however casually offered

3. Support responsible, low-impact ecotourism at national parks and sanctuaries

4. Participate in citizen-science initiatives like bird counts or camera-trap surveys

5. Create pollinator-friendly spaces with native plants where possible

6. Support local afforestation and habitat restoration efforts

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **Project Tiger's Recovery Story:** India's tiger population grew from a crisis low of 1,411 in 2006 to 3,682 in 2023, making India home to an estimated 70–75% of the world's wild tigers — one of the most cited wildlife recovery successes globally.
- **Kaziranga National Park (Assam):** A UNESCO World Heritage Site with the highest tiger density of any protected area in the world, and home to around 70% of the global population of the greater one-horned rhinoceros.
- **Sundarbans National Park (West Bengal):** The world's largest mangrove forest and a UNESCO World Heritage Site, uniquely home to tigers adapted to swimming and hunting in brackish tidal waters.
- **International Big Cat Alliance (IBCA):** Launched by India in 2023 and formally established as a treaty-based intergovernmental body in January 2025, bringing together 27 countries to drive global big cat conservation.

Practical Scenarios

Scenario 1: While travelling, you see wildlife products (ivory, animal skins, or exotic pets) being sold at a local market.

Do not purchase these items. Note the location and details if safe to do so, and report it to the nearest forest department office or the wildlife crime control helpline.

Scenario 2: A wild animal has strayed into agricultural land near a village bordering a forest.

Avoid direct confrontation, alert the local forest department promptly, and support community-based deterrent measures rather than retaliatory harm to the animal.

Scenario 3: You are planning a visit to a national park or wildlife sanctuary.

Choose a registered, responsible tour operator, stay within designated zones, avoid feeding wildlife, and maintain a respectful distance to minimise disturbance to natural behaviour.

Government Rules / Guidelines

- **Ministry of Environment, Forest and Climate Change (MoEFCC):** The apex national body for wildlife and forest policy, protected area management, and India's international conservation commitments.
- **National Tiger Conservation Authority (NTCA):** Oversees Project Tiger, advises states on tiger reserve declarations, and coordinates the All India Tiger Estimation conducted every four years.
- **Wildlife Institute of India (WII):** A premier research institution supporting wildlife science, training, and the National Wildlife Information System.
- **State Forest Departments:** Implement day-to-day protection, patrolling, and management of protected areas at the state level.

Laws & Policies

Law / Policy	What It Covers
Wildlife Protection Act, 1972	Protects wild animals, birds, and plants; regulates hunting and trade; establishes protected area categories
Forest (Conservation) Act, 1980	Regulates diversion of forest land for non-forest use
Biological Diversity Act, 2002	Protects India's biological resources and traditional knowledge from unauthorised use
Project Tiger, 1973	India's flagship tiger conservation programme, now integrated with Project Elephant under NTCA
CITES (International)	Regulates international trade in endangered species, which India is a signatory to
Environment (Protection) Act, 1986	The umbrella law empowering government action against activities that harm the environment, including habitats

> *All Tiger Reserves in India are governed under the Wildlife (Protection) Act, 1972, and administered by the National Tiger Conservation Authority (NTCA), which advises state governments on reserve declarations and coordinates protection efforts nationally.*

Important Statistics

Metric	Figure
Total protected areas in India (2023)	1,014 (106 National Parks, 573 Wildlife Sanctuaries, 115 Conservation Reserves, 220 Community Reserves)
Share of India's geographical area under protection	About 5.32%
India's wild tiger population (2023 estimate)	3,682 (up from 1,411 in 2006)
India's share of the world's wild tiger population	Approximately 70–75%
Number of Tiger Reserves under Project Tiger (2025)	58 (up from 9 in 1973)
Great Indian Bustard population (critically endangered)	Approximately 100–150 individuals remaining
India's plant species as a share of the world total	About 7%
Central allocation for wildlife habitats (FY 2025–26)	■450 crore, including ■290 crore for Project Tiger and Elephant

(Figures sourced from the Ministry of Environment, Forest and Climate Change (MoEFCC), the National Tiger Conservation Authority (NTCA), and Press Information Bureau releases; official portals should be checked for the latest updates.)

Best Practices

- Report suspected poaching or illegal wildlife trade promptly
- Choose responsible, registered operators for wildlife tourism
- Avoid purchasing any products made from wildlife parts
- Support local afforestation and habitat restoration initiatives
- Maintain safe, respectful distance from wildlife during any encounter
- Participate in citizen-science wildlife monitoring where available

Common Mistakes

- Feeding or approaching wild animals at parks and sanctuaries
- Purchasing souvenirs made from wildlife parts, even unknowingly
- Ignoring or not reporting suspected wildlife crime
- Assuming a single conservation success (like tigers) means all species are equally safe
- Supporting unregulated, high-disturbance wildlife tourism

Myths vs Facts

Myth	Fact
"India's wildlife is generally safe now that tiger numbers have recovered"	Species like the Great Indian Bustard remain critically endangered with fewer than 150 individuals left, showing conservation success is species-specific
"Protected areas are only for tourists and researchers"	Protected areas conserve essential ecosystem services like water regulation and pollination that benefit surrounding communities directly
"Human-wildlife conflict means wildlife populations are becoming too large"	Conflict is largely driven by habitat fragmentation and shrinking natural range, not overpopulation
"Reporting wildlife crime is not an individual citizen's responsibility"	Citizen reporting is often the first alert that enables forest departments to act against poaching and illegal trade

Do's and Don'ts

Do	Don't
Report suspected poaching or wildlife crime	Don't purchase products made from wildlife parts
Choose responsible, registered tour operators	Don't feed or approach wild animals
Support local habitat restoration efforts	Don't support unregulated wildlife tourism
Maintain safe distance during wildlife encounters	Don't assume all species are equally safe from decline
Participate in citizen-science monitoring	Don't ignore signs of illegal wildlife trade

Practical Tips

For Travellers & Tourists

Book only registered, responsible safari operators • Keep a safe, respectful distance from wildlife • Avoid loud noise or flash photography near animals

For Communities Near Protected Areas

Support eco-development and alternative livelihood programmes • Use community-based early warning systems for wildlife movement • Coordinate with forest departments on conflict mitigation

For Schools & Families

Encourage participation in bird counts and nature walks • Create native, pollinator-friendly plants at home or school • Discuss the difference between conservation and captivity

Practical Checklist

■	I know the difference between a national park, sanctuary, and biosphere reserve
■	I know which authority to contact to report suspected wildlife crime
■	I avoid purchasing any products made from wildlife parts
■	I choose responsible, registered operators for wildlife tourism
■	I maintain safe distance and avoid disturbing wildlife during encounters
■	I know at least one biodiversity hotspot located in or near my region
■	I have discussed wildlife conservation with family or peers

Quick Revision Notes

- Biodiversity = variety of life at genetic, species, and ecosystem levels
- India is a megadiverse country with four global biodiversity hotspots
- Protected area categories: national park, wildlife sanctuary, conservation reserve, community reserve, biosphere reserve
- India's tiger population recovered from 1,411 (2006) to 3,682 (2023) under Project Tiger
- Key laws: Wildlife Protection Act 1972, Forest (Conservation) Act 1980, Biological Diversity Act 2002
- NTCA, MoEFCC, and state forest departments are India's key wildlife regulatory bodies

Key Takeaways

1. India's biodiversity is globally significant — decisions made here affect species found nowhere else on Earth

2. Conservation success varies dramatically by species — tiger recovery does not mean every species is safe

3. Protected areas exist on a spectrum, from strict national parks to community-managed reserves

4. Citizen reporting and responsible tourism directly support enforcement and habitat protection

5. Human-wildlife conflict is primarily a habitat and space issue, not a sign of wildlife overabundance

Glossary

- **Biodiversity Hotspot** — A region with exceptional species richness that is also significantly threatened
- **Buffer Zone** — The surrounding area of a protected reserve allowing regulated sustainable use
- **Core Zone** — The strictly protected inner area of a tiger reserve
- **Endemic Species** — A species found naturally in only one specific geographic region
- **Habitat Fragmentation** — The breaking up of large, continuous habitats into smaller, isolated patches
- **IUCN Red List** — The global inventory of conservation status for species
- **Keystone Species** — A species with a disproportionately large effect on its ecosystem
- **Megadiverse Country** — One of a small group of nations home to the majority of the world's species
- **Poaching** — The illegal hunting or capturing of wild animals
- **Wildlife Corridor** — A protected passage linking separated habitats for safe animal movement

Frequently Asked Questions

1. What is biodiversity?
The variety of life on Earth at genetic, species, and ecosystem levels — encompassing all plants, animals, fungi, and microorganisms and the ecological communities they form.

2. Why is India considered a megadiverse country?
India is one of 17 globally recognised megadiverse nations, home to nearly 7% of the world's plant species and four internationally recognised biodiversity hotspots.

3. What is the difference between a national park and a wildlife sanctuary?
National parks offer the highest protection level with no human settlement or resource extraction allowed; wildlife sanctuaries protect specific species or habitats while permitting some regulated human activity.

4. How has India's tiger population changed over time?
It fell to a crisis low of 1,411 in 2006, then recovered to 3,682 by 2023 under Project Tiger, making India home to an estimated 70–75% of the world's wild tigers.

5. What is Project Tiger?
India's flagship tiger conservation programme, launched in 1973 with 9 reserves and now covering 58 reserves, administered by the National Tiger Conservation Authority.

6. What is a biodiversity hotspot?
A region with exceptional species richness that is also significantly threatened by habitat loss; India has four: the Himalayas, Western Ghats, Indo-Burma region, and Sundaland.

7. Which law is India's primary wildlife protection legislation?
The Wildlife Protection Act, 1972, which bans hunting of protected species and establishes India's protected area categories.

8. What is human-wildlife conflict, and what causes it?
Negative interactions between people and wild animals, usually arising from habitat fragmentation and shrinking natural range rather than wildlife overpopulation.

9. Is the Great Indian Bustard still endangered despite tiger recovery?
Yes. It remains critically endangered, with an estimated 100–150 individuals left in the wild, largely due to collisions with power lines.

10. How can I report suspected wildlife crime?
Contact your local State Forest Department or use the Wildlife Crime Control Bureau's reporting channels under the Ministry of Environment, Forest and Climate Change.

Official References

- Ministry of Environment, Forest and Climate Change (MoEFCC) — protected area and wildlife policy information
- National Tiger Conservation Authority (NTCA) — All India Tiger Estimation reports
- Wildlife Institute of India (WII) — National Wildlife Information System
- Press Information Bureau (PIB) — releases on India's wildlife conservation milestones
- Wildlife Protection Act, 1972; Forest (Conservation) Act, 1980; Biological Diversity Act, 2002 — India Code portal

Official Government Website Links

- Ministry of Environment, Forest and Climate Change: **https://moefcc.gov.in**
- National Tiger Conservation Authority: **https://ntca.gov.in**
- Wildlife Institute of India: **https://wii.gov.in**
- Project Tiger / NTCA Portal: **https://projecttiger.nic.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

India's biodiversity is not a backdrop to daily life — it is a living inheritance, shaped by decades of conservation effort and now increasingly shaped by the choices of ordinary citizens. The tiger's recovery shows what sustained, science-led conservation can achieve; the Great Indian Bustard's decline shows how much remains fragile. Every reported violation, every responsible tourism choice, every supported restoration effort adds to the balance.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay curious, stay protective, and remember: what we choose to conserve today is what future generations will inherit.

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